

Computational modelling of strut defects in SLM manufactured lattice structures

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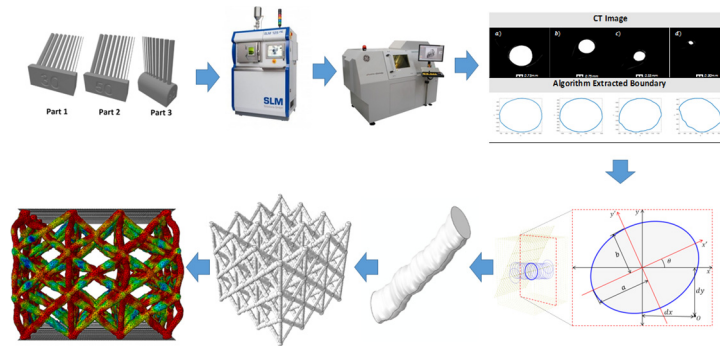
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HIGHLIGHTS

- A computationally inexpensive method of generating AM representative lattice geometries for FE modelling is proposed.
- The method is successfully applied to predict the mechanical response of SLM fabricated Inconel 625.
- Extracted geometrical properties can form the basis of a database for numerical modelling of AM lattice topologies.
- Geometrical data can be used to randomly generate AM representative strut models for stochastic FEM.

GRAPHICAL ABSTRACT



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ABSTRACT

Addition of manufacturing defects to computational models of 3D-printed lattice structures enable improved simulation accuracy. A computational model of a cellular structure based on finite element method (FEM) analysis, often starts from defect-free computer-aided design (CAD) geometries to generate discretised meshes. Such idealised CAD geometries neglect imperfections, which occur during the additive manufacturing process of lattice structures, resulting in model oversimplification. This research aims to incorporate manufacturing defects in the strut elements of a lattice structure, thereby enhancing predictive capabilities of models. In this work, a method of generating CAD AM representative strut models is proposed. The models are generated from micro-computer tomography (μ CT) analysis of SLM fabricated struts. The proposed additive manufacturing (AM) representative strut FE model's axial stiffness and critical buckling load is compared to idealised- and μ CT- based FE models, with significant error reduction over idealised strut models. The AM representative strut models are then used to generate full lattice FE models and compared with manufactured and idealised FE models. The AM representative FE lattice models show greater correlations toward experiment and more realistic deformation behaviours. Overall, the methodology used in this study demonstrates a novel approach to representing struts in FE models of AM fabricated lattice structures.

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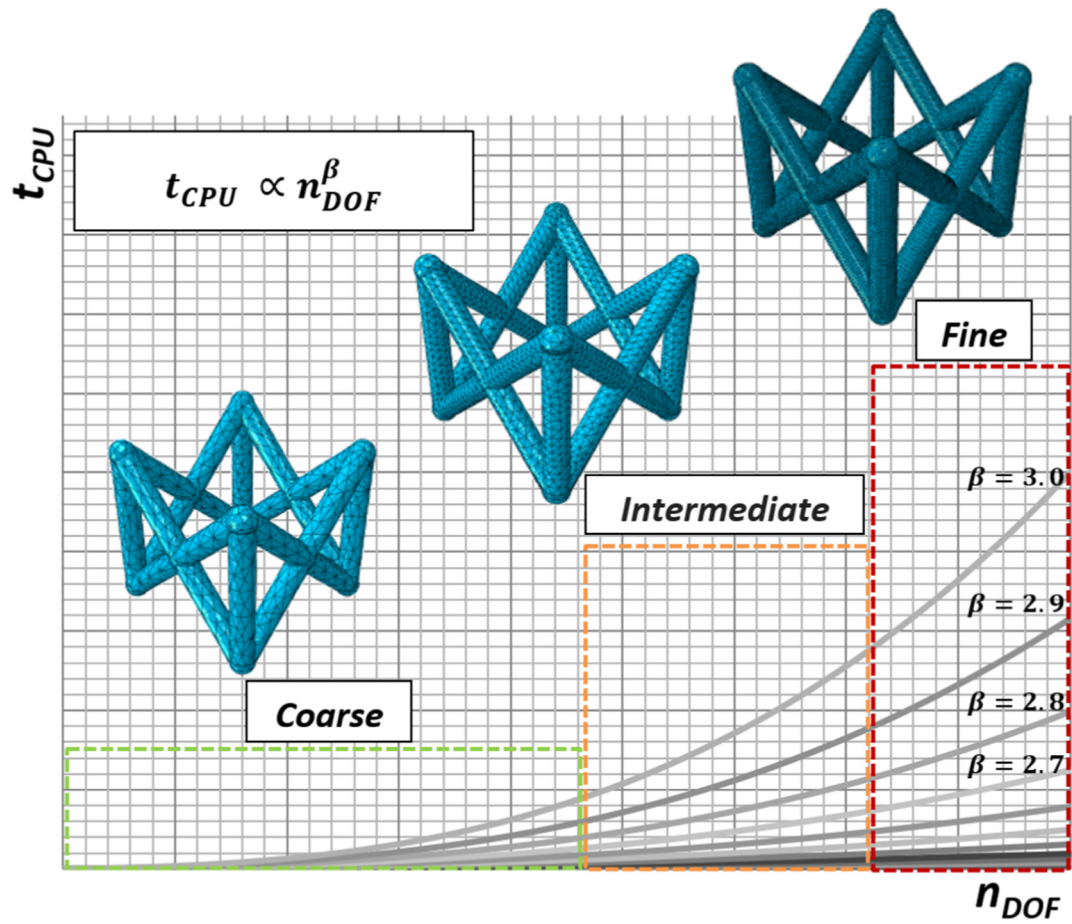
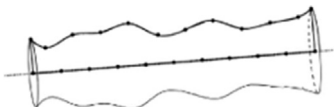
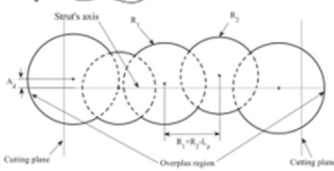
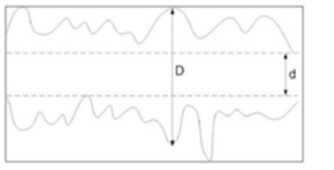
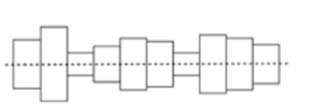


Fig. 1. Schematic of relationship between CPU time, t_{CPU} , required to solve a finite element model with specified DOF, n_{DOF} . Representative images of coarse, intermediate and fine mesh densities are shown. The exponent, β , is a constant typically between 2 and 3 for linear elastic FE models.

Table 1

Methods of generating strut defects within CAD and FE models of AM lattice structures for subsequent simulation. Including analysis of methods attributes.

METHOD		ATTRIBUTES					
		Simulate experimentally derived minimum strut diameter	Variable strut diameter	Capture stress concentrations along strut length	Uni-directional centroid deviation	Bi-directional centroid deviation	Simulate varying cross-section shape
   	Spline of n points defines a body of revolution with respect to an axis [58].	✓	✓	✓	✗	✗	✗
	Boolean of N number of spheres creating entire structure [37].	✓	✓	✓	✓	✓	✗
	Evaluation of maximum cross section distance, d between parallel lines that fall inside strut [62].	✓	✗	✗	✗	✗	✗
	Beam element model generated with various cross-sectional diameters [61]	✓	✓	✓	✗	✗	✗

1. Introduction

Additive manufacturing (AM) refers to fabrication *via* successive addition of material to form a specific part defined by 3D model data [1]. The addition of material to form parts is unique to AM which differentiates it from traditional *subtractive* and *formative* manufacturing processes [2].

AM utilizing a metallic material can be further classified as a metal additive manufacturing (MAM) process. MAM processes permit the design and manufacture of geometrically complex lattice structures from metals and alloys such as: titanium [3], Inconel [4] and aluminium [5]. Lattice structures display high structural efficiency [6] and can be economically fabricated at low production volumes [7] with MAM; having found commercial application in high-value aerospace and patient-specific orthopaedics.

However, there can exist large variations in the performance of these structures as MAM processes tend to produce geometric defects in the fabricated lattice. There is lack of design tools which capture these defects and enable analysis of local deformation behaviours *via* numerical models. The outcomes of this research aim to: provide a novel method of accommodating AM associated defects within CAD lattice models thereby enhancing the predictive capabilities of applied numerical models.

1.1. Metallic additive manufacturing (MAM) of lattice structures

Selective laser- and electron beam melting (SLM and EBM, respectively) are powder bed fusion (PBF) based MAM technologies [8], involving part fabrication *via* programmed delivery of an energy source (laser or electron beam) to the surface of a metallic powder bed [9]. Fabrication of the solid three-dimensional part is a repetitive process, consisting of: melting of a layer of the components cross-sectional geometry on the metallic powder bed, followed by lowering of the bed (*via* the build platform). The bed's surface is then recoated with a powder layer of specified thickness, for melting of the next cross-sectional slice [10]. Gibson *et al* [11] simplified PBF associated processing parameters into four interdependent parameter groups: laser, scan, powder and temperature.

SLM offers an attractive method for the fabrication of geometrically complex lattice structures. The available literature identifies some of the following research contributions associated with MAM of lattice

structures, their applications, mechanical properties and numerical models:

- Mazur *et al* [12] analysed the compressive behaviour of SLM manufactured titanium (Ti-6Al-4V) lattice structures with varying unit cell topologies, strut diameters, cell sizes and bulk geometries. Solid linear-elastic FE models were also created to qualitatively compare the simulated mechanical response of the various unit cell topologies to experiments.
- Leary *et al* [5] studied the compressive behaviour of SLM manufactured AlSi12Mg lattice structures with varying unit cell topologies, strut diameters, cell sizes and bulk geometries. Solid linear-elastic FE models were constructed using micro- computer tomography (μ CT) reconstructions of the fabricated structures for comparison of numerical models with experiments.
- Yanez *et al* [13] analysed the compressive behaviour of sixteen different EBM manufactured Ti6Al4V gyroid structures, the main variable for differentiating each structure was the angle that its struts made with the axial direction.
- Sing *et al* [14] reviewed manufacturability and mechanical testing considerations for Ti-6Al-4 V, stainless steel 316 L and cobalt-chromium (Co—Cr) SLM fabricated lattice structures. Exhibiting the dependency of manufacturability on the selected unit cell topology.
- Li *et al* [15] performed compression tests on SLM fabricated stainless steel 316 L lattice structures with body-centred cubic (BCC) unit cell topology. A dynamic explicit FE model was created to compare with experimental compression tests, including a fabricated strut derived elastoplastic material model.

Strut-level defects generated during the MAM (*via* PBF processes) of lattice structures impact the mechanical properties of the fabricated part and therefore the ability of numerical models to predict them [23]. Ravari *et al* [24] observed strut-specific defects caused by the SLM process which influence its mechanical behaviour, including: variation in cross-sectional geometry along the strut length, the 'waviness' of the strut and micro-porosity. Waviness describes the deviation of the strut's axis across its length. This factor combined with a variable cross-sectional diameter along the strut's length cause local

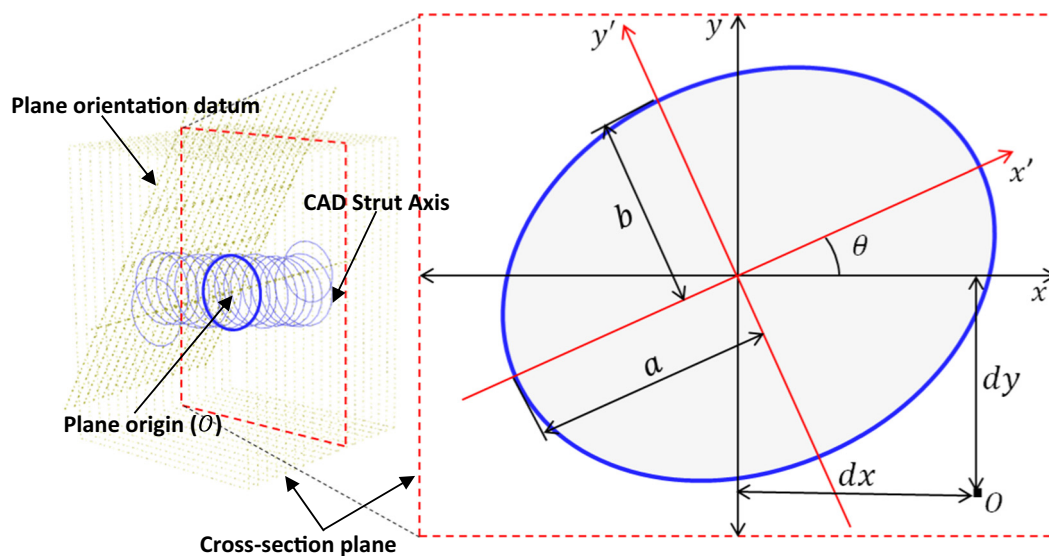


Fig. 2. Representation of elliptical strut cross-sections used for loft operation in modelling of lattice structures and normal view of cross-sections showing variables used to define ellipse, where: a , b , dx , dy , x' and y' are: major axis diameter (derived from I1), minor axis diameter (derived from I2), x centroid relative to theoretical strut axis (plane origin, O), y centroid position relative to theoretical strut axis (plane origin), x' and y' are principal axis of inertia defined by axis orientation angle θ .

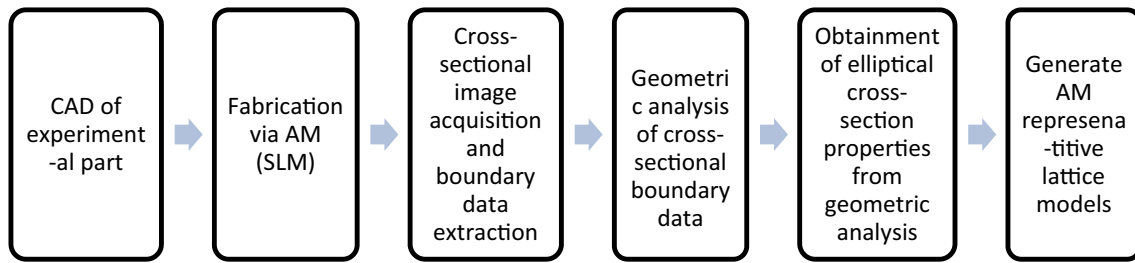


Fig. 3. Overview of method used to generate data to specify strut defects within CAD and FE models of struts and lattice structures.

heterogeneities and stress concentrations leading to stiffness and compressive strength variability [25].

Weißmann *et al* [26] studied the fabrication quality of SLM and EBM manufactured Ti-6Al-4 V struts and its influence on the struts mechanical properties including the variation between target CAD cross-sectional diameter and the as-manufactured diameter. Van Bael *et al* [25] utilised μ CT scanning to investigate morphological differences between the designed face- and body-centred cubic (FBC) lattice structures and the SLM manufactured Ti-6Al-4 V part. It was observed that the deviation from CAD strut thickness to as-manufactured one was greater than 100% for struts of 100- μ m diameter. Sing *et al* [27] used scanning electron microscopy (SEM) to show differences between the CAD and fabricated struts within commercially pure titanium (CP-Ti) SLM built lattice structures. Their work showed the effect of partially and fully adhered metal particles to the surface of the strut.

1.2. Deformation behaviour of lattice structures

The deformation behaviour of these lattice structures can be categorised as *bending-* or *stretch-dominated* [16]. Mechanically, *stretch-dominated* behaviour accompanies high-strength, low-compliance structures; conversely, *bending-dominated* structures are associated with relatively low-strength and highly-compliant structures [17]. Unit cells in *bending-dominated* lattices deform via generation of bending moments at strut intersections and subsequent formation of plastic-hinges (ductile materials). Whilst in *stretch-dominated* structures, tensile or compressive forces are generated in struts, with plastic deformation and failure governed by yielding, buckling or the brittle collapse of struts. *Bending-* or *stretch-dominated* lattice structures could be utilised to design smart lattice devices exhibiting complex behaviours under different types of loadings [18,19].

Application of traditional force methods and Maxwell's stability criterion is a simple predictive method enabling insight into the expected characteristics of a particular lattice structure [17,20,21]. However, analyses of local stress concentrations such as strut-level manufacturing

defects and material non-linearity require more complex numerical or analytical methods to be employed [22].

1.3. Numerical modelling of lattice structures

Utilisation of numerical analysis, specifically the finite element method (FEM), for lattice-structure design can assist the selection of material and geometrical factors for optimal performances (high strength-to-stiffness ratio). Finite-element (FE) models are typically constructed using idealised computer-aided design (CAD) geometries. AM often induces geometric defects caused by the manufacturing process, leading to performance discrepancies between simplified models and the as-fabricated components. The severity and type of these geometric defects is a function of the associated MAM technique, specific process parameters and local geometries. Consequently, the use of idealised CAD geometries to model an AM lattice's structural response oversimplifies the fabricated structures geometry and thereby reduces the models' predictive capabilities. There are several approaches presented in literature to simulate AM lattice associated manufacturing defects, which include: varying diameter beam element cross-sections, joining of various sized spheres emulating a meandering strut axis and the CAD revolution of a spline about an axis to represent a struts geometry in three-dimensions.

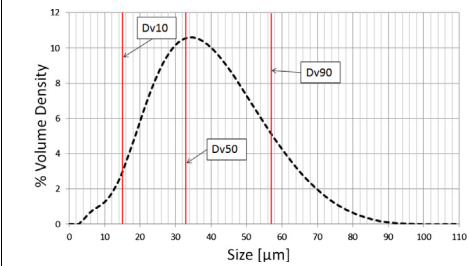
Defect inclusion methods are further detailed in Section 2.3. These methods are limited due to their focuses on variations in lattice's strut diameter along the length and deviation of the struts' centroid along the axis, while neglecting the change in the strut's cross-sectional shape. This research proposes a novel method of generating CAD lattice models to incorporate micro-scale geometric defects. The proposed method for generating AM representative lattice models is based on mimicking geometric information found in fabricated struts. Section 2 of this work is a brief literature review of both CAD and FE modelling of lattice structures. It examines techniques for the CAD generation of lattices, FE modelling of lattices undergoing compression and the inclusion of AM manufacturing defects. Section 3 covers: the

Table 2
Part design for the analysis of manufacturing defects.

CAD Model		Part	Strut Axis – Build Platform Inclination Angle (°)	Strut Diameters (mm)
	Part 1	1	35.6	0.1 to 1.0
	Part 2	2	45	0.1 to 1.0
	Part 3	3	90	0.1 to 1.0

Table 3

SLM Processing parameters and utilised Inconel 625 powder size distribution.

Particle Size Distribution	SLM Parameter	Value
	Layer thickness (μm)	30
	Focal offset (μm)	0
	Laser power (W)	200
	Energy density (J/mm^3)	61.7

experimental design for analysis of AM lattice defects, the proposed method of including those defects within CAD lattice models and the numerical modelling approach. Section 4 presents the results from numerical models with Section 5 concluding the findings and applications of this research.

2. Overview of modelling approaches for MAM lattice structures

CAD and FE modelling are often used to assist designing lattice structures and enabling visualisation, fabrication and to capture mechanical responses. The following section examines methods and considerations to ensure robust CAD design and FE modelling of lattice structures for AM.

2.1. CAD modelling

Strut's spatial arrangement within a lattice structure can be stochastic or periodic affected by design and fabrication process. In this work, lattice structures, which are constructed from periodic unit cells, will be investigated numerically taking into account the geometrical variations induced by AM. In recent work, Azman *et al* [28] included various design considerations such as: selection of unit cell topology, surface limitations, progressivity and conformity of the unit cell for constructing periodic lattice structures to fill in a design volume.

The generation of CAD models during lattice structure design is an important step for effective visualisation, AM pre-processing and model import into FEA software [29,30], as well as conversion to stereolithography (STL) file format for AM fabrication [31]. Methods of generating CAD models for periodic lattice structures, which conform to specified geometric bounds include exploitation of FEM element discretisation to generate the conformal lattice arrangements [32,33] and the application of Boolean methods [34,35]. Programmatic methods of STL lattice generation have also been proposed by McMillan *et al* [36].

2.2. Finite element modelling

This work aims to model the quasi-static compression of lattice structures in which local plasticity occurs at low global strain [37]. Therefore, a non-linear static FE model is developed as the inertial and damping forces in the system are negligible.

The selection of a FE model's element type impacts simulation outcome. Both beam and continuum FE models have been developed to simulate lattice structures' mechanical behaviour [38]. Typically, beam models help to reduce the computational cost, whilst solid models allow for a more accurate depiction of local mechanical behaviour [38] as well as the capturing the influences of geometric defects in three-dimensions.

Reducing computational cost can be challenging during simulation of a lattice structures' mechanical responses. FE models of highly porous

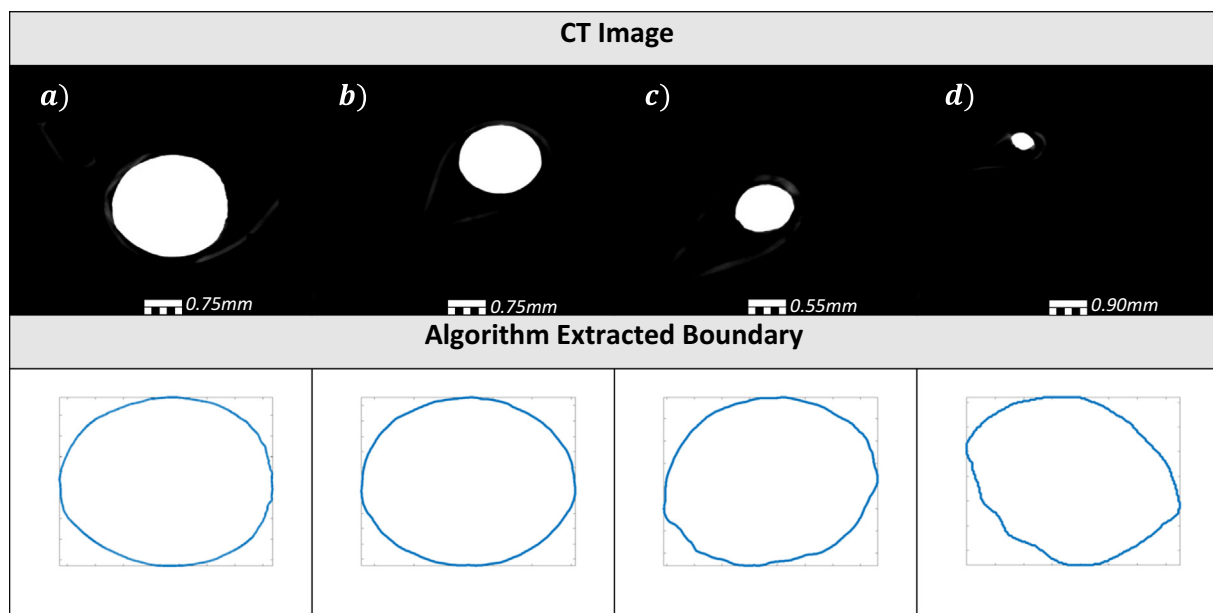


Fig. 4. Examples of raw CT cross-sectional images and their extracted boundary, for: a) 3 mm diameter at 35.6° inclination angle, b) 2 mm diameter at 90° inclination angle, c) 1 mm diameter at 35.6° inclination angle and d) 0.5 mm diameter at 45° inclination angle.

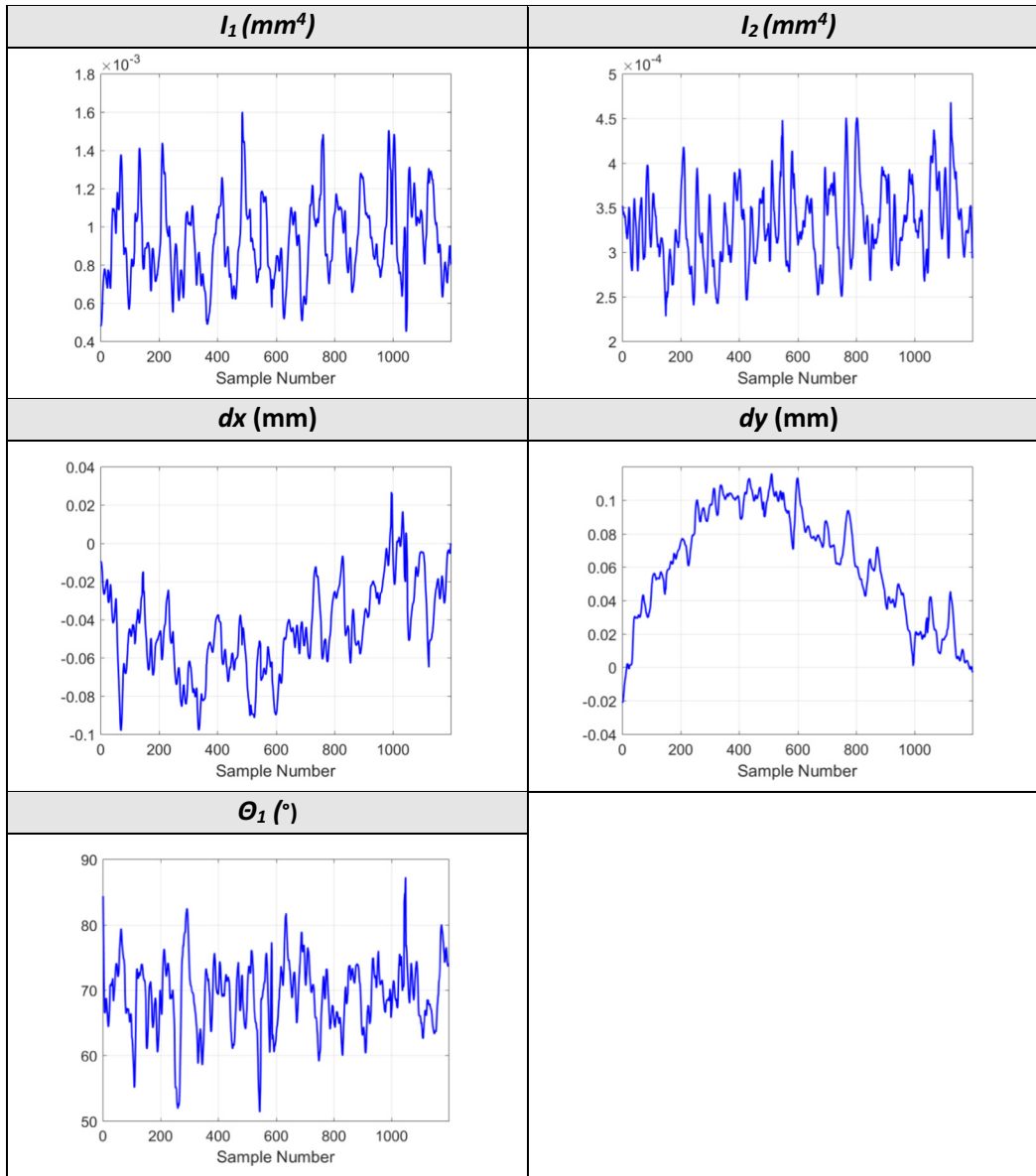


Fig. 5. Cross-sectional properties extracted for a 300 μm diameter strut at 35.6-degree inclination angle.

macro-scale lattice structures containing multiple high aspect-ratio micro/meso-scale struts require an extremely fine 3D FE mesh. Such fine FE 3D meshes are associated with significant number of degrees of freedom (DOF) in the system and therefore computationally intensive. Generally, for static analysis CPU time can be related by [39]:

$$t_{\text{CPU}} \propto n_{\text{DOF}}^\beta \quad (1)$$

where, t_{CPU} is the CPU time, n_{DOF} is the total number of degrees of freedom and β is a constant ranging from 2 to 3 (Fig. 1). Therefore, a large porous lattice structures would require extensive computational resource and infeasible FE models, requiring a reduction the model size [40,41] and limitation for *mesh convergence studies* [42]. Tran et al. [43,44] have developed simplified beam model for lattice unit cell to capture the global behaviour of the structure. Their proposed model also included the imperfections of lattice structure by introducing the weighted disturbances in different deformation modes.

On the other hand, FE analysts, which use computationally intensive three-dimensional elements to predict the mechanical behaviours of lattice structures can benefit from ability to incorporating the

manufacturing defects. An accurate depiction of the as-manufactured lattice part can improve the models overall predictive capabilities.

2.3. Inclusion of MAM defects in FE models

Various methods of introducing defects into computational models of AM lattice structures have been proposed (Table 1). A relatively straightforward method of defect inclusion is the utilisation of μCT scanning of fabricated lattices to generate geometries for FE models. Xiao et al [45] recommended using this method after large disparities between idealised FE models and experiments in their work, also concluding that the introduction of random imperfections in the idealised geometries may also increase accuracy.

Gümürük and Mines [46] compared beam and solid FE models with experiments, to predict the compressive behaviour of SLM fabricated stainless steel 316 L lattice structures. Discrepancies between beam element models and experimental data are attributed to beam models' inability to capture as-manufactured strut intersections, model the variation in strut diameter and residual semi-melted powders on the structure.

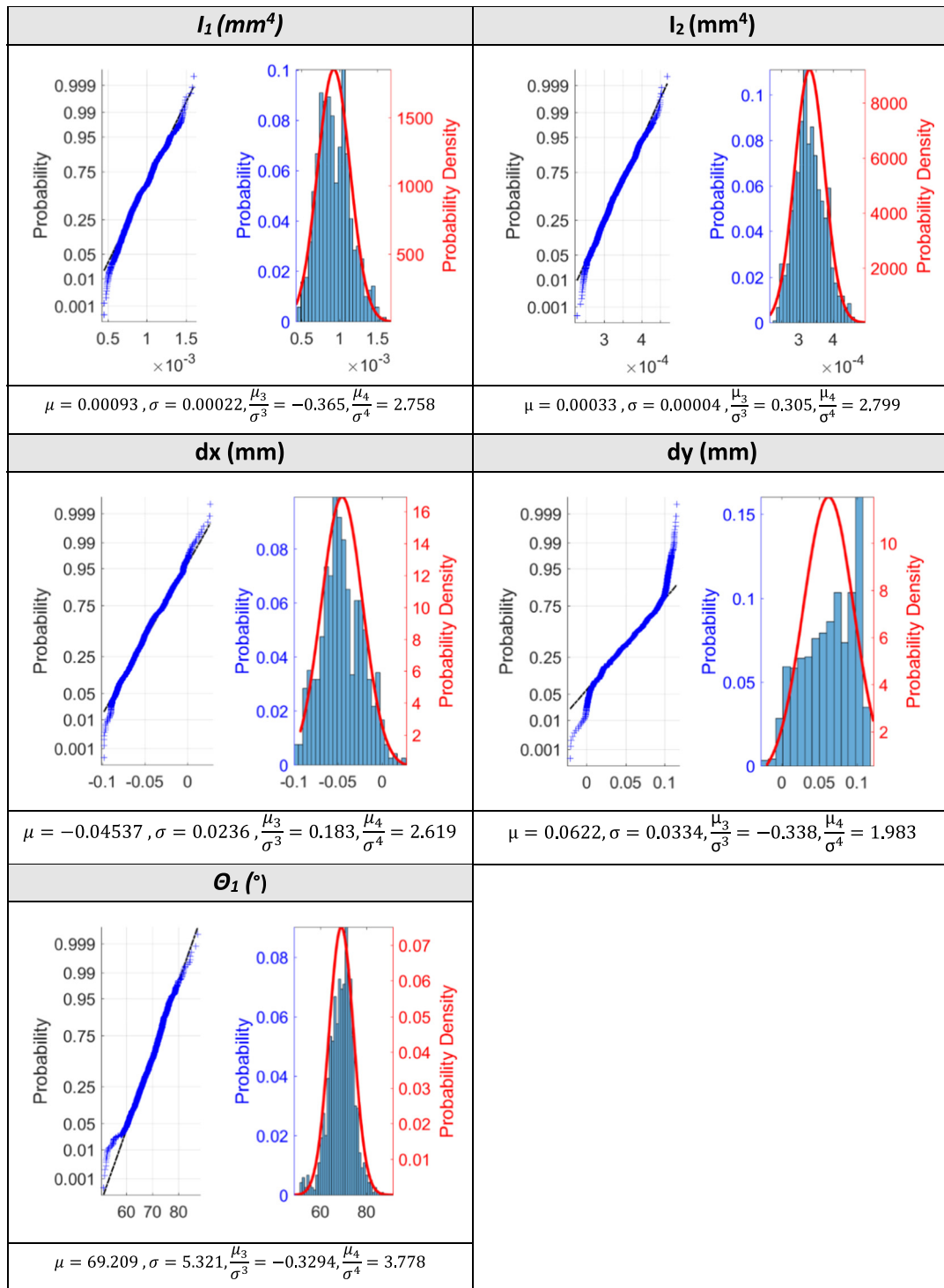


Fig. 6. Probability plot and histogram with probability density outlined in red, for 300 μm diameter strut at 35.6-degree inclination angle. Including values of: Mean (μ), standard deviation (σ), skewness (μ_3/σ^3) and kurtosis (μ_4/σ^4).

Ravari *et al* [47] included the variation of a strut's diameter in a body-centred cubic lattice with Z-struts (BCCZ) fabricated via fused deposition modelling (FDM). Defect struts were modelled via the revolution of a spline about the struts intended axis forming the solid CAD geometry. The spline was defined by a number of points, where each point is representative of struts radii relative to the axis of revolution. The applied modelling method introduces variation in the strut's diameter along its length. However, other defects such as deviations from strut's axis and change in cross-sectional shape cannot be modelled using this technique.

Ravari and Kадkhodaei [48] proposed a method for modelling variations in a fabricated strut's diameter and deviation of its axis (waviness). Tsopanos *et al* [49] included defects based on microstructural observations of SLM manufactured stainless steel 316 L lattice structures. The proposed method of strut defect inclusion consisted of merging of spheres with varying diameters and centroidal positions relative to the strut's axis. This defect generation approach is unable depict change in cross-sectional shape.

Campoli *et al* [50] modelled variation of strut diameter using beam elements with varying specified diameters, which were drawn from a

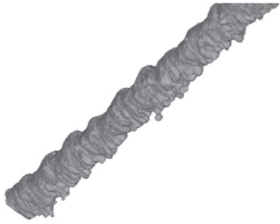
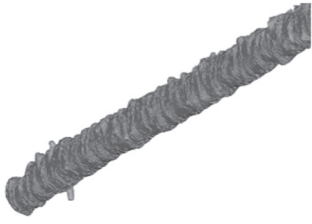
CT scans		
300 micron @ 35.6 degrees	300 micron @ 45 degrees	300 micron @ 90 degrees
		
AM representative CAD geometry		
300 micron @ 35.6 degrees	300 micron @ 45 degrees	300 micron @ 90 degrees
		

Fig. 7. μ CT reconstructions and their AM representative CAD models for 300- μ m diameter struts built at: a) 35.6°, b) 45° and c) 90° to the build platform.

Gaussian distribution of SEM determined diameters. Smith *et al* [38] compared beam and solid FE models of single BCC and BCCZ unit cells to experimental results from a multi-cell lattice structure. Yang *et al* [51] compared analytical and numerical models to experimental results for a Ti-6Al-4V EBM manufactured lattice structure with a 3D re-entrant

unit cell topology. Modelled struts were uniform in diameter and constructed from the minimum cross-sectional diameter determined via SEM. This method could be useful for critical analysis lattice structure strength.

3. Detailed model development

Overall, inclusion of geometric defects in FE models of AM lattice structures has been limited to using CAD models with a minimum strut diameter found experimentally. The modelling of a varying diameter along its length and models capturing ‘waviness’ of a strut. Reviewed methods (Table 1) neglect important geometrical properties of the struts cross-sectional area, which impact the mechanical behaviour of a strut. The proposed method will aim to address above-mentioned limitations by accounting for an increased number of geometric features, which are determined from fabricated parts. Section 3.1 covers the proposed CAD modelling technique for generating AM representative strut models, Sections 3.2 and 3.3 present the experimental design used to obtain the geometric properties of fabricated struts. Section 3.4 displays the resultant AM representative strut CAD models and their use in the construction of AM representative lattice

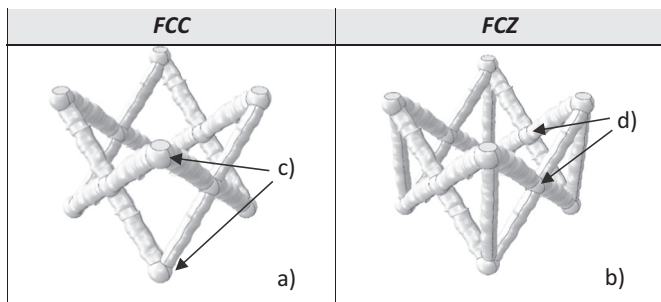


Fig. 8. AM representative CAD models for a) FCC and b) FCZ unit cells with 300- μ m strut diameters. Images also show examples of c) boundary connection spheres and d) connection spheres, which are used for load application and Boolean merge operations.

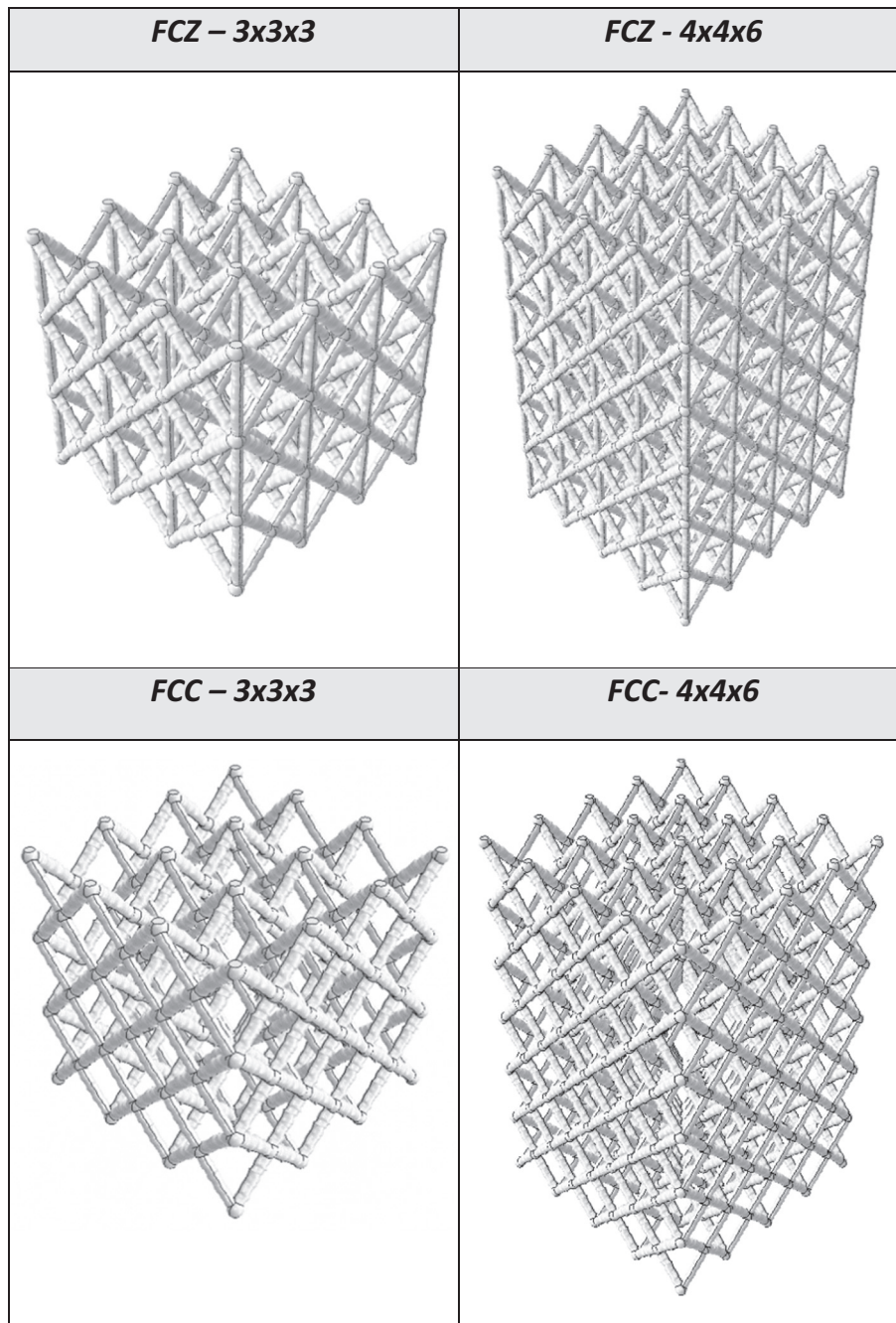


Fig. 9. AM representative CAD models for FCZ and FCC lattice structures of $3 \times 3 \times 3$ and $4 \times 4 \times 6$ sizes.

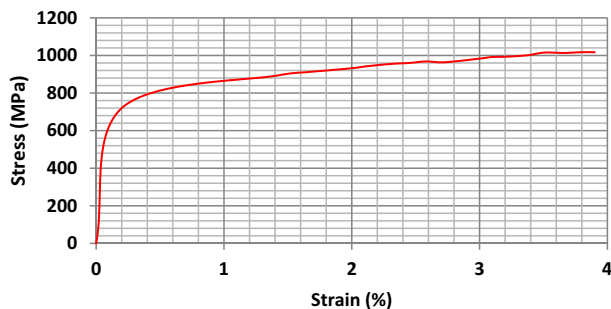


Fig. 10. Experimental test data for SLM-manufactured Inconel 625 [52].

models. Section 3.5 covers the construction of a FE model to simulate compression of a lattice using the AM representative geometries generated.

3.1. AM representative CAD strut models

To account for manufacturing defects on a strut's structural properties, and to include these changes in CAD reflecting as-manufactured cross-sectional shape, an elliptical cross-section is proposed in this work. This cross-section profile will be used instead of idealised circular cross-section, which is quite popular in CAD lattice models. Elliptical cross-sections offer the ability to mimic the varying principal area moments of inertia and the associated principal axis orientation of strut's cross-section. This can be achieved by matching the ellipses' major and minor axis: radii and planar orientation, to micro-CT (μ CT) derived

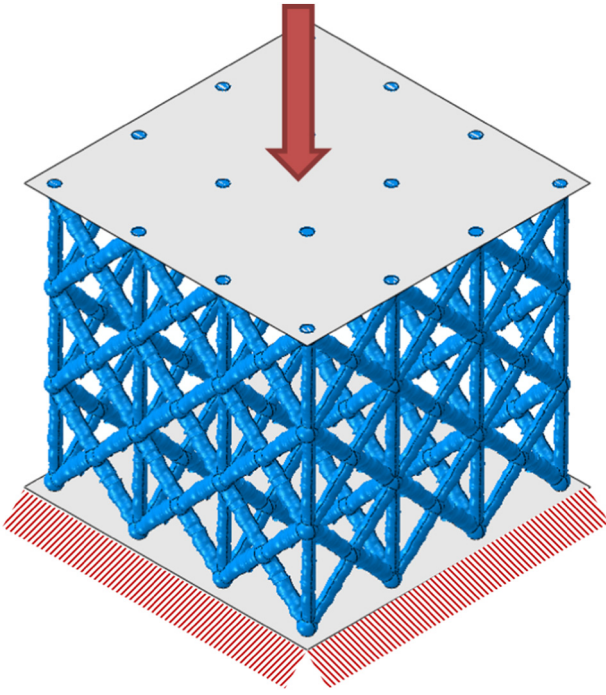


Fig. 11. Schematic representation of FE model with AM representative lattice geometry. Also showing rigid plates used for compression analysis and loads/boundary conditions.

principal area moments of inertia and associated principal axes orientation.

Modelling AM representative three-dimensional solid strut geometries from elliptical cross-sections is achieved via CAD loft operations, where several cross-sections are specified along the axis of the strut constructing its shape. The elliptical cross sections are joined via a single smooth path passing through each cross-section's centroid.

The set of two-dimensional planes defining the struts shape have origins, which are co-linear in three-dimensional space and located on the strut's axis. Therefore, accounting for deviation in a fabricated struts axis is achieved by simply translating the defined ellipse away with the plane's origin (Fig. 2).

3.2. Design of experiment

The method for generating data to specify an elliptical cross-sections properties involved μ CT scanning of SLM fabricated struts, extraction of cross-sectional images and subsequent image processing and analysis (Fig. 3).

3.2.1. CAD of experimental part

Three parts were designed (Table 2) enabling analysis of strut manufacturing defects, these parts consisted of a base region with 10 protruding struts of 0.1 to 1 mm (in 0.1 mm steps) diameters and 10 mm lengths. To examine the effect of build inclination angle (angle between struts axis and build platform) each part had its struts modelled at a specific inclination angle of 90, 45 or 35.6 degrees (Table 2). The selected inclination angles represent those found during the manufacture of lattice structures with BCC and FCC unit cell topologies as well as Z-strut variants.

3.2.2. Part fabrication via SLM

All parts were fabricated via SLM from nickel-based superalloy Inconel 625, due to its high ductility and resistance to fracture in comparison to typical medical-grade MAM materials [4,12]. SLM Solutions 250 machines were used with processing parameters shown in Table 3. The powder size distribution is represented by particle

diameters (D_v) and their associated volume densities (Table 3). Evaluated 10th, 50th and 90th percentiles (D_{v10} , D_{v50} and D_{v90}) diameters were 14.7, 33 and 57.1 μ m, respectively. The moment mean's for surface area ($D_{[2,3]}$) and volume/mass ($D_{[3,4]}$) were found to be 25.2 and 34.7 μ m, respectively.

3.2.3. Cross-sectional image acquisition and boundary data extraction

Custom holding devices are fabricated via Fused Deposition Modelling (FDM) enabling consistent strut orientation during μ CT scanning² each part. Scans were performed with a 7.5 μ m voxel size, followed by thresholding and three-dimensional reconstruction. A binary image stack of cross-sectional views was generated at 8- μ m intervals. A custom developed algorithm is used to analyse the image stack and extract scaled cross-sectional boundary data. Fig. 4 displays the scripts capability.

3.2.4. Geometric analysis of cross-sectional boundary data

The cross-sectional information extracted from the boundary data included centroid position Cartesian co-ordinates (X, Y), centroidal principal area moments of inertia (I_1 and I_2) and the principal axis inclination angle (θ_1 and θ_2). Data from each sectional property, which lie outside three standard deviations from the mean was removed. Therefore, for a μ CT cross-sectional image to be deemed 'valid' its sectional properties all must lie within three standard deviations from the mean of the associated property distribution.

The principal moments of inertia, $I_1, 2$, were obtained via numerical calculation as:

$$I_{1,2} = \frac{I_{xx} + I_{yy}}{2} \pm \sqrt{\left(\frac{I_{xx} - I_{yy}}{2}\right)^2 + I_{xy}^2} \quad (2)$$

where I_{xx} and I_{yy} , represent moments of inertia and I_{xy} as the product of inertia. The angle of inclination of the principal axis, $\theta_{1,2}$, is evaluated as:

$$\tan 2\theta_p = \frac{-I_{xy}}{2(I_x - I_y)} \quad (3)$$

The two roots of θ_p give the inclination angles of the rotated axes in a counter-clockwise direction from the horizontal axis (x-axis).

3.2.5. Obtaining elliptical cross-section properties from geometric analysis

Utilisation of an elliptical cross-section to mimic the as-manufactured struts cross-section required specification of a major and minor axis radii which is based upon extracted principal moment of inertia data. This was achieved by constructing a system of equations based upon the moment of inertia for an ellipse and equating them to the experimental ones:

$$\begin{cases} I_1 = I_{xx} = \frac{\pi}{4} ab^3 \\ I_2 = I_{yy} = \frac{\pi}{4} a^3 b \end{cases} \quad (4)$$

The ellipses radii a and b are obtained from Eq. (4) as:

$$a = \frac{4}{\pi b^3} I_1 \quad (5)$$

$$b = \left(\frac{16 \cdot I_1^3}{\pi^2 \cdot I_2} \right)^{\frac{1}{8}} \quad (6)$$

The ellipses orientation is defined by its principal axis x' and y' . The x' principal axis is rotated in a counter-clockwise direction by calculated

² Scanned via Bruker Skyscan 1275

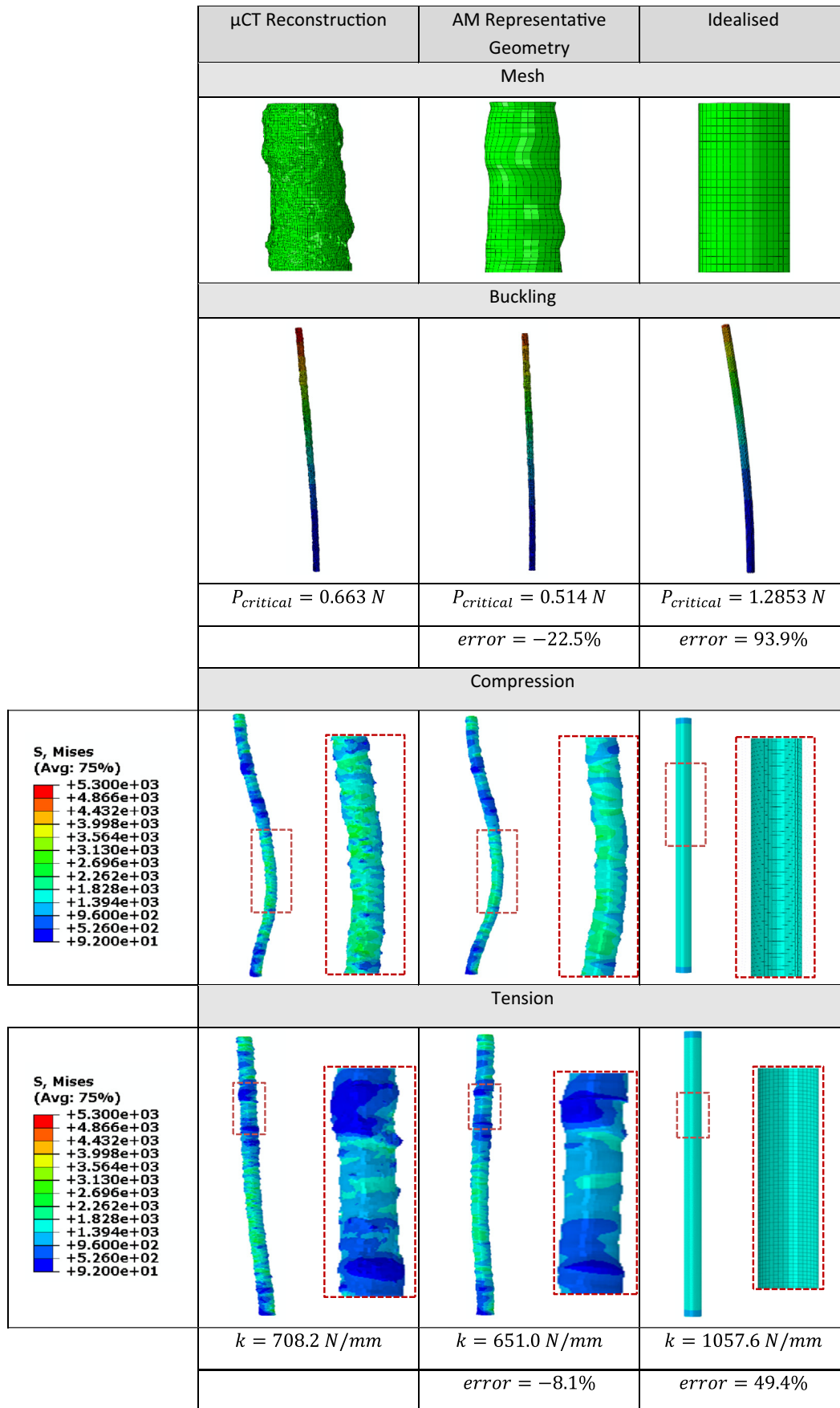


Fig. 12. Finite element mesh of μ CT reconstructed, AM representative and idealised strut geometries with simulated buckling, tension and compression deformed shapes. Evaluated stiffness, k , and critical buckling load, $P_{critical}$, are also displayed and their error relative to the μ CT reconstruction. Stress is displayed in MPa.

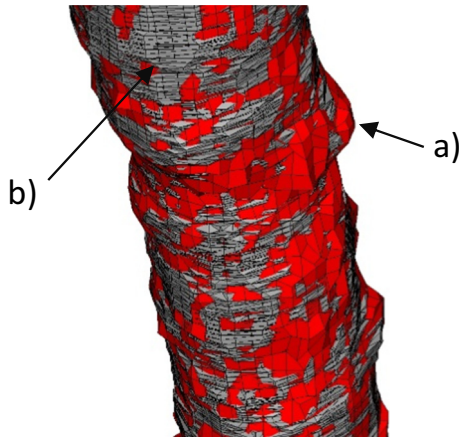


Fig. 13. Finite element mesh (red) of μ CT reconstructed strut (grey) showing accuracy of applied shrink wrap meshing technique. Where: a) FE mesh region larger than μ CT reconstruction and b) μ CT reconstruction region larger than its representative FE mesh.

angle, θ_1 . The y' principal axis rotation angle is calculated by:

$$\theta_2 = \theta_1 + \frac{\pi}{2} \quad (7)$$

The inclination angles themselves are kept to between 0 and 180° for simplicity.

The deviation in the fabricated struts centroid along its length is evaluated by first selecting boundary data from the start and the end positions of the image stack (which is ordered based upon its location in the strut). A linear regression line is then fit between the associated start and end X centroid position, and its location along the strut. Centroid deviation, dx , is equal to the difference between the regression

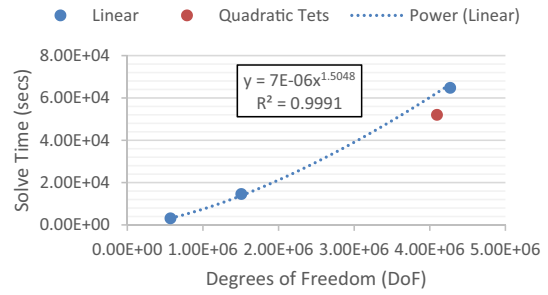


Fig. 15. Simulation solve time (t_{CPU}) versus total number nodal degrees of freedom (n_{DOF}).

predicted and observed data. The described process is repeated for Y centroidal position to obtain centroid deviation, dy .

3.3. Resultant cross-sectional properties

Extracting cross-sectional profiles from the fabricated parts using methods described in Section 3.2 is similar to sampling a continuous signal of strut cross-sectional shapes at a frequency of approximately 120 samples per mm (Fig. 5 and Fig. 6). Therefore, when generating the representative CAD model for AM strut, an equivalent elliptical cross-section is defined at the same frequency.

3.4. AM representative CAD lattice models

Face-centred and face-centred cubic with Z-struts (FCC and FCZ, respectively) CAD lattices were generated (Fig. 9) with AM representative strut geometries (Fig. 7). Idealised lattice models is also generated for finite element analysis and contain the same strut geometry used in the stereolithography input file for AM (i.e. undistorted cylindrical struts).

Connection spheres are modelled at strut intersections with a diameter 20% larger than the target struts diameter (Fig. 8), allowing effective Boolean merge operations and the formation of a lattice

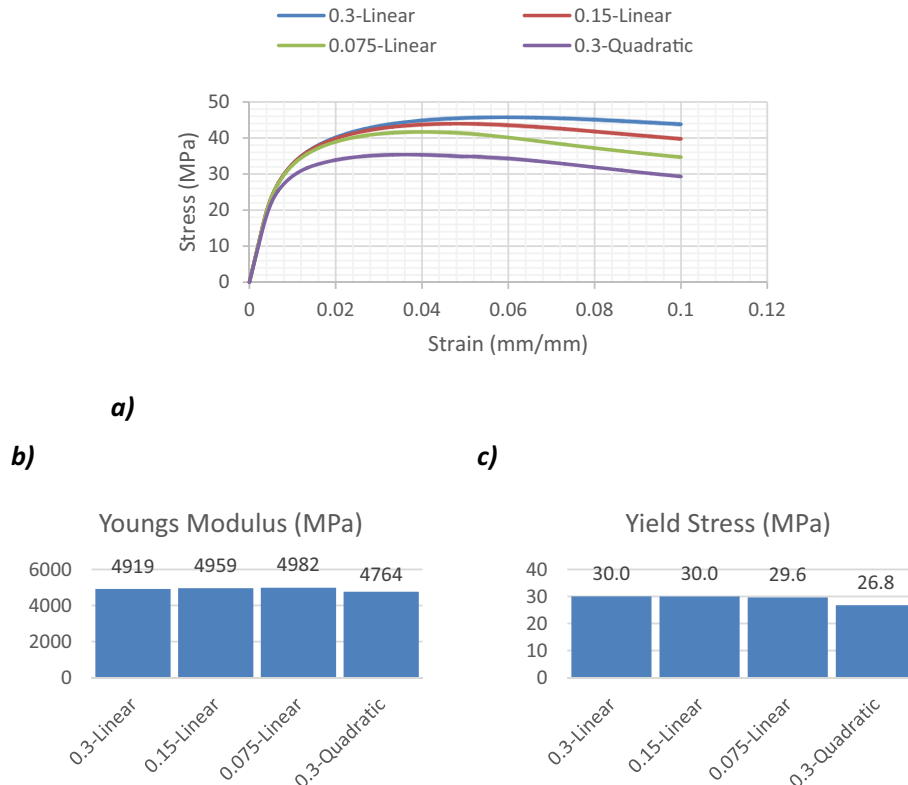


Fig. 14. Results from mesh convergence test for every element size and order tested, where: a) Stress-strain curve, b) Young's Modulus and c) Yield Stress.

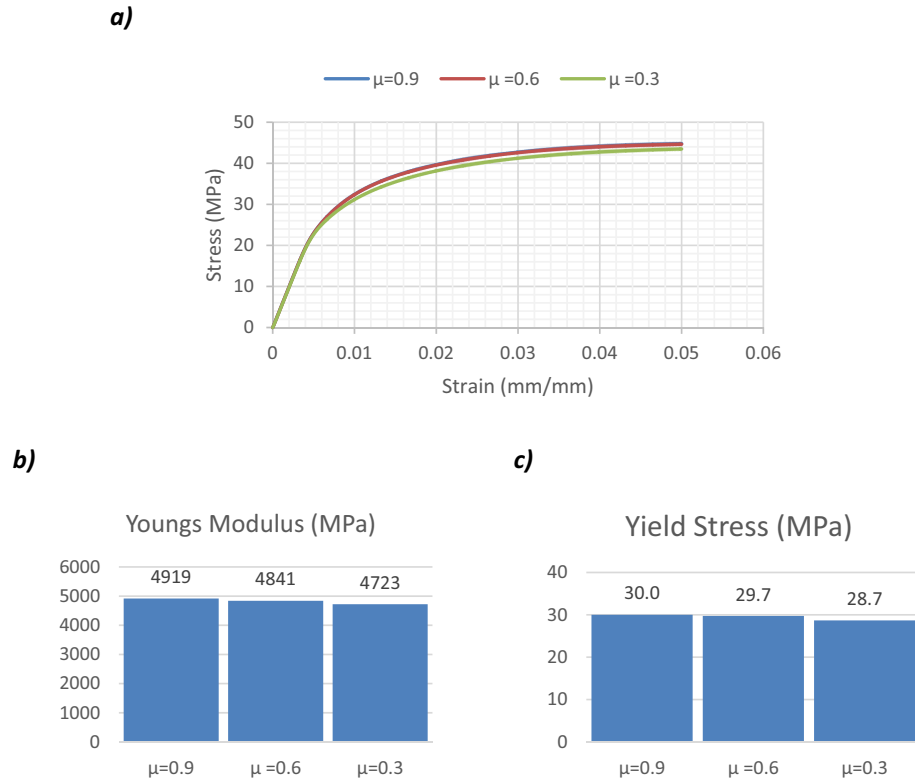


Fig. 16. Results from sensitivity analysis of the plate-lattice interface: a) Stress-strain curve, b) Young's Modulus and c) 0.2% Yield Stress.

geometry. Boundary connection spheres are located at contact points or boundary conditions (uppermost and lowermost sphere rows) and are modelled as spherical caps (Fig. 8). The flat portion of the spherical cap is oriented with a normal direction parallel to the applied load. The height of the removed cap equalled a quarter of the sphere's diameter.

3.5. Finite element model construction

Simulating the generated AM representative lattice geometries undergoing quasi-static compression is achieved by constructing a non-linear static FE model. The material model is based on *Yadroitsev et al* [52] work on the mechanical properties of SLM fabricated Inconel 625. Material properties is extracted from mechanical tests carried out on samples, which are loaded parallel to the build direction. The material properties defined in the simulation includes a Young's modulus of 140 GPa, Poisson's ratio of 0.308, density of 7980 kg/m³ and a tabular set of flow stresses and associated true plastic strains (Fig. 10).

Rigid plates are used to model the compression of lattice structures, which is represented by surface-to-surface contact. Behaviours of the plate-lattice contact in the normal direction is set to a 'hard' type relationship which minimises plate penetration and disallows transfer of tensile stress across the interface [30]. A friction sensitivity analysis is performed in Section 4.2 to understand the effect of the prescribed tangential frictional coefficient at the interface, resulting in a selected coefficient of 0.9.

All degrees of freedom were constrained on the lower plate, analogous to a stationary platen during compression testing. The upper rigid plate is prescribed a displacement which equals a 5% compressive strain (Fig. 11). Both plates are modelled to be in contact with the lattice core from the beginning. Resultant mechanical properties are derived from a reference point on the upper rigid plate, which is rigidly coupled with all plate's nodes.

Due to the geometric complexity of the lattice structures being modelled, linear and quadratic tetrahedral elements are used (C3D4 and C3D8, respectively). A mesh convergence study is performed in

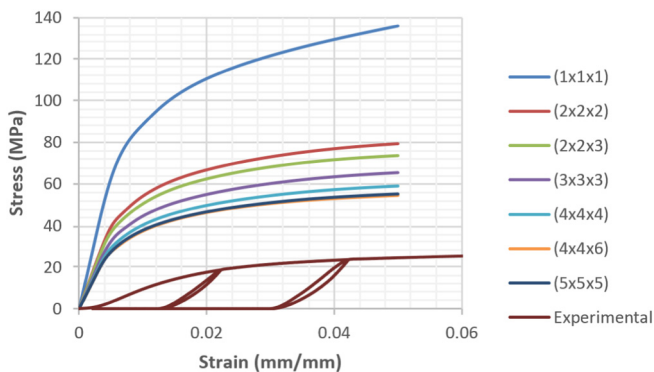


Fig. 17. Stress-strain curve extracted from finite element analysis of an FCZ idealised lattice structure undergoing compression, for a range of lattice sizes.

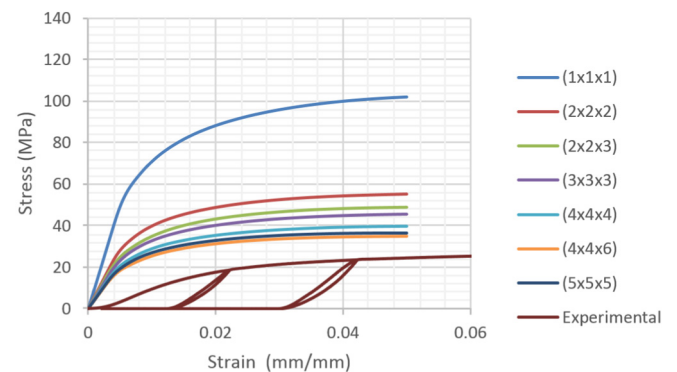


Fig. 18. Stress-strain curve extracted from finite element analysis of an FCZ AM representative lattice structure undergoing compression, for a range of lattice sizes.

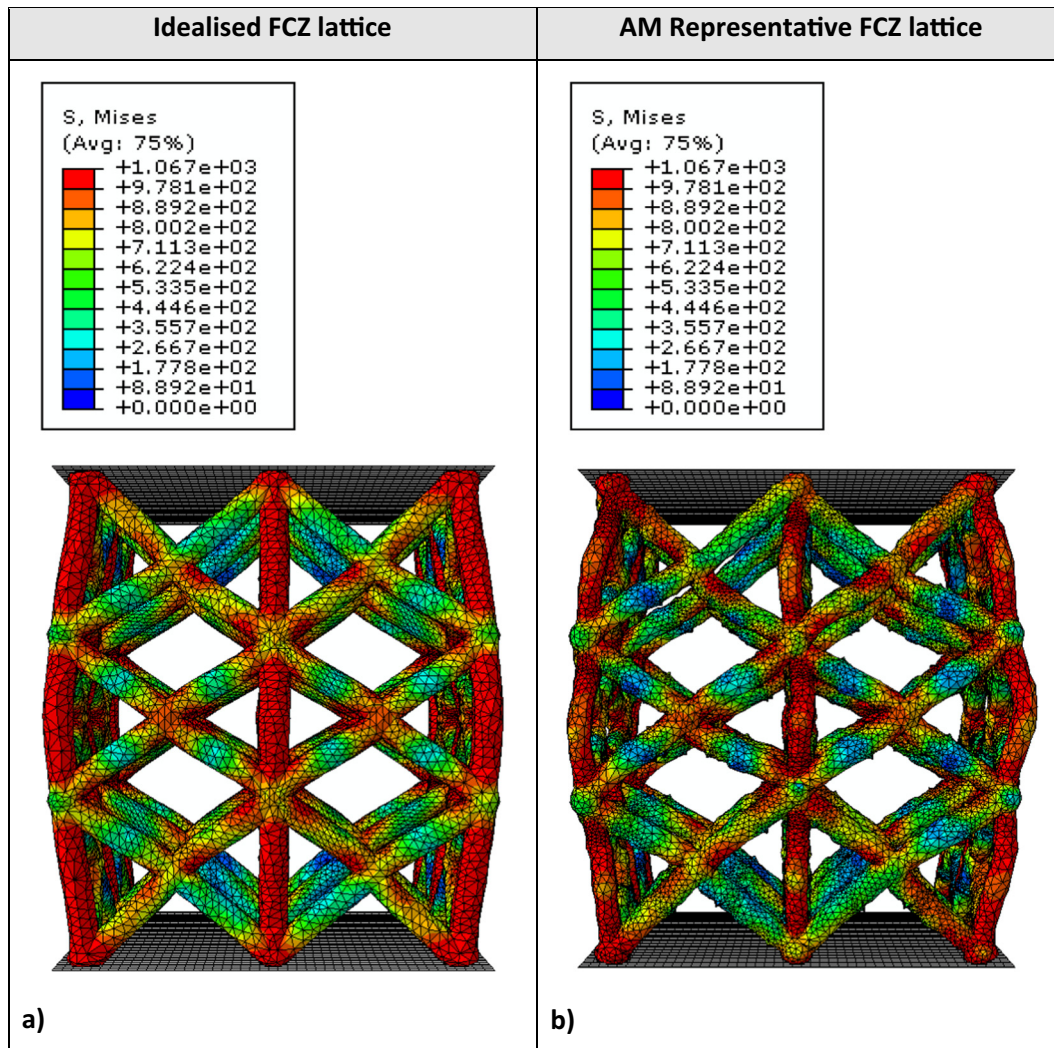


Fig. 19. Equal strain case Von-Mises stress distribution for $3 \times 3 \times 3$ FCZ lattice with a) idealised geometry and b) AM representative geometry. The stress is displayed in MPa.

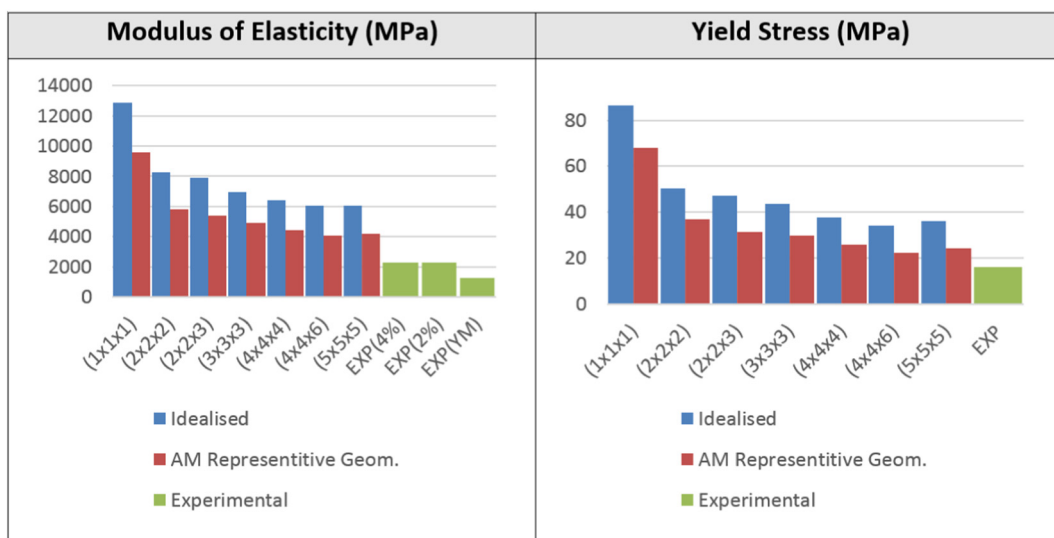


Fig. 20. Modulus of elasticity and yield stress for simulated idealised and AM representative FCZ lattice structures, over a range structure sizes. Also shown are experimentally derived: Young's modulus, 2% and 4% unloading moduli, as well as yield stress.

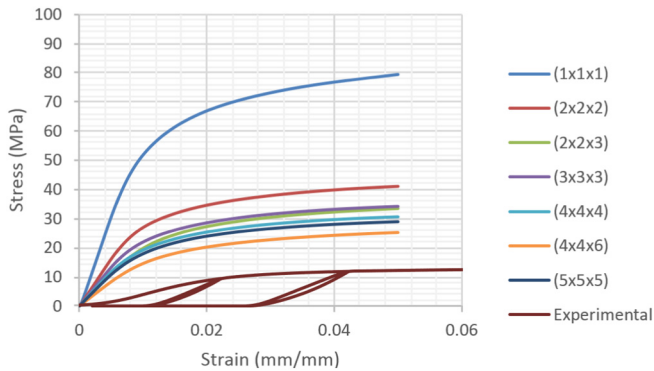


Fig. 21. Stress-strain curve extracted from finite element analysis of an FCC idealised lattice structure undergoing compression, for a range of lattice sizes.

Section 4.2 to select the final element size and type, which balances computational cost and accuracy.

4. Numerical results and discussions

The following sections presents the results and discussion in accordance with the approach prescribed in the previous numerical model section.

4.1. Comparisons between μ CT reconstructed and idealised strut models

To verify the proposed method of generating AM representative strut models, linear buckling, linear-elastic compression and tension FE models are created. Models are based on the μ CT reconstruction, AM representative and idealised strut geometries. A strut with a 300- μ m diameter, 90° build inclination angle and 9.33 mm length is chosen.

The buckling analysis consisted of a fixed-free boundary condition arrangement with an axial unit load applied to the free end. Critical buckling load ($P_{critical}$) is evaluated at the first buckling eigenmode and used for comparison. For compression and tension analysis, a 1% strain is applied in the strut's axial direction via a reference node rigidly coupled to the strut's uppermost surface. Strut stiffness (k) is calculated from the reaction force at the reference node. A shrink-wrapping approach was used to create a FE mesh from the μ CT reconstruction (Fig. 13). All analysis used first order hexahedral elements and the material model consisted of the same Young's Modulus and Poisson's ratio presented in Section 4.2.

Results from the buckling, tension and compression analyses are presented in Fig. 12, with stiffness and critical buckling load error compared to the μ CT reconstruction model. Both analyses show an increase

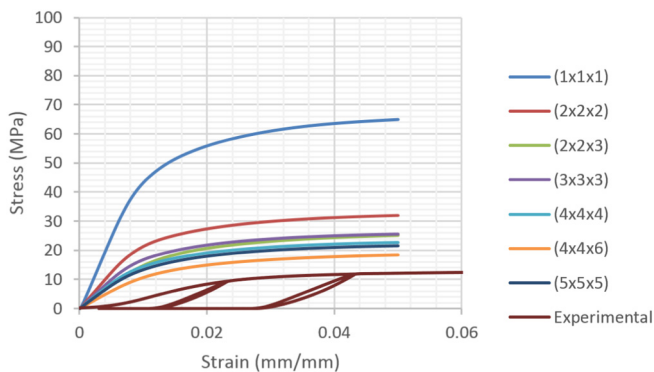


Fig. 22. Stress-strain curve extracted from finite element analysis of an FCC AM representative lattice structure undergoing compression, for a range of lattice sizes.

in accuracy when utilizing an AM representative strut geometry. The idealised geometry over-predicts stiffness and critical buckling load of the μ CT reconstructed strut, and conversely, the AM representative strut under-predicts stiffness and critical buckling load.

Error in stiffness and critical buckling load between the AM representative geometry and the μ CT reconstruction models (Fig. 12) can most likely be attributed to the applied shrink-wrapping method.

As visible in Fig. 12, the FE mesh protrudes the boundary of the μ CT reconstruction. This leads to regions of increased strut diameter, which may cause the greater stiffness and critical buckling load seen in the μ CT reconstruction model in comparison to the AM representative geometry.

4.2. Lattice FE model verification

A global element size reduction (*h-refinement*) and element order increase (*p-refinement*) are the mesh refinement strategies for lattice model. Convergence metrics include Young's Modulus and 0.2% Yield Strength. The Young's Modulus is defined by the slope of a regression line fit to the linear portion of the stress-strain curves. Mesh refinement is performed on a $3 \times 3 \times 3$ FCZ AM representative lattice undergoing a 10% compressive strain and the plate-lattice interface is set to have friction coefficient of 0.9.

The *h-refinement* mesh study prescribes an initial element edge length of 0.3 mm and halved it for each refinement iteration. A total of 3 iterations are completed with prescribed element edge lengths of 0.3, 0.15 and 0.075 mm. The *p-refinement* component of the study held element edge length at 0.3 mm, while element could be linear or quadratic (see also Fig. 14).

The first reduction in element size (*h-refinement*) produces a 0.81% and 0.06% difference in Young's Modulus and yield stress, respectively. The second element size reduction produces a difference of 0.46% and 1.30%. An increase in element order (*p-refinement*) produced a 3.2% and 11.26% difference in Young's Modulus and Yield Stress, respectively. Simulations exhibits a relatively considerable difference when element order is increased, though negligible when compared with the significant increase in computational time (Fig. 15). Therefore, first order elements are selected with a target element edge length of 0.3 mm for all simulations.

A sensitivity analysis of the prescribed tangential frictional coefficient (plate-lattice interface) is performed for frictional coefficients of 0.9, 0.6 and 0.3.

The analysis (Fig. 16) indicates a negligible difference in predicted Young's Modulus and 0.2% Yield Stress. A friction coefficient of 0.9 is selected for the remainder of simulations due to its higher predicted stiffness and yield stress.

4.3. Effect of lattice size on mechanical properties

Experimental data is obtained from Leary et al. [4] on the mechanical properties of Inconel 625 lattice structures. The parts fabricated in this work (Section 3.2) are fabricated on the same machine, with the same processing parameters and powder feedstock. The experimental lattices are equivalent to those simulated, with target strut diameters, unit cell topologies and cell sizes. The number of cell repetitions could not be matched in simulations due to the high computational cost. Therefore, the simulated lattices are restricted to those in Table 4.

4.3.1. FCZ lattice FE models

Stress-strain plots for the idealised and AM representative FCZ lattices are presented in Fig. 17 and Fig. 18, respectively, including the experimental curve. Both idealised and AM representative lattices exhibit converging behaviour of their mechanical properties, this size convergence behaviour is also seen in work by Hedayati et al [53]. Visual comparison of the idealised and AM representative stress-strain plots show the AM representative structures converging toward the experimentally derived curve more accurately.

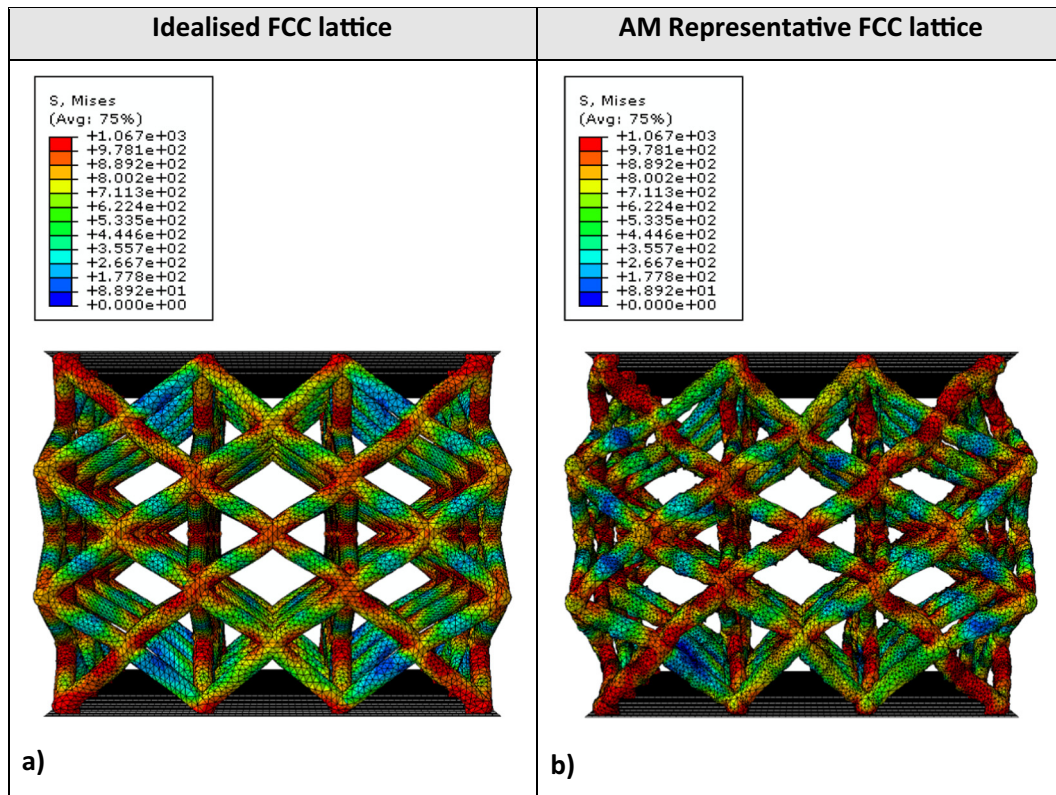


Fig. 23. Equal strain case Von-Mises stress distribution for $3 \times 3 \times 3$ FCC lattice with a) idealised geometry and b) AM representative geometry. The stress is display in MPa.

The deformed lattice shape and Von Mises stress distribution (Fig. 19) for the AM representative geometry displays increased realism in strut deformation behaviour. Struts within the deformed AM representative structure exhibits failure in a buckling-dominated manor, in comparison to a crushing-like failure seen in the idealised lattice geometry.

Fig. 20 presents the Young's moduli and yield stress for both the idealised and AM representative FE models. The Young's modulus tends to decrease with an increased number of unit cell repetitions, this behaviour is also seen with the predicted yield stress. For both metrics the AM representative geometry predicts a structure which is more compliant and less strong then its idealised counterpart.

The experimentally derived values of Young's Modulus are effected by: compression platen angle, support removal and localized plasticity below yielding. These effects are not accounted for in lattice structure simulation, which can lead to over-stiff predictions of the Young's modulus.

Ashby *et al* [54] recommended measuring the Young's Modulus of metallic cellular structures by the slope of the unloading curve as it better represents the structures performance. Therefore, simulations should be compared with the unloading moduli (Fig. 20, EXP (2%) and EXP (4%)). For the idealised and AM representative $4 \times 4 \times 6$ lattice a 90% and 56% difference with experimental 2% unloading modulus is

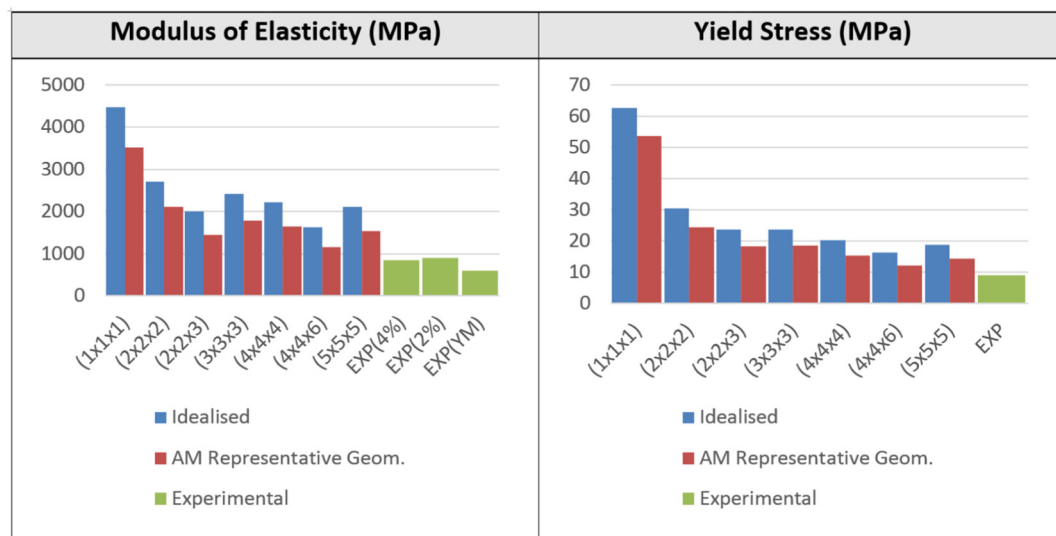


Fig. 24. Modulus of elasticity and yield stress for simulated idealised and AM representative FCC lattice structures, over a range structure sizes. Also shown are experimentally derived: Young's modulus, 2% and 4% unloading moduli, as well as yield stress.

Table 4

Overview of simulated and experimental lattice structures.

Topology	Cell size (mm)	Strut diam. (mm)	Num. of cells (X)	Num. of cells (Y)	Num. of cells (Z)	Applied test
FCZ	2	0.3	1	1	1	Simulated 5% strain
FCZ	2	0.3	2	2	2	Simulated 5% strain
FCZ	2	0.3	2	2	3	Simulated 5% strain
FCZ	2	0.3	3	3	3	Simulated 5% strain
FCZ	2	0.3	4	4	4	Simulated 5% strain
FCZ	2	0.3	4	4	6	Simulated 5% strain
FCZ	2	0.3	5	5	5	Simulated 5% strain
FCZ	2	0.3	10	10	15	Experimental compression
FCC	2	0.3	1	1	1	Simulated 5% strain
FCC	2	0.3	2	2	2	Simulated 5% strain
FCC	2	0.3	2	2	3	Simulated 5% strain
FCC	2	0.3	3	3	3	Simulated 5% strain
FCC	2	0.3	4	4	4	Simulated 5% strain
FCC	2	0.3	4	4	6	Simulated 5% strain
FCC	2	0.3	5	5	5	Simulated 5% strain
FCC	2	0.3	10	10	15	Experimental compression

seen, respectively. Simulated yield stress performed in a similar manner as the Young's Modulus with size converging behaviour exhibited.

The experimental lattice's base length and width to height ratio (AR) was 2:3, this aspect ratio is mimicked with the simulation of a $2 \times 2 \times 3$ and $4 \times 4 \times 6$ lattices. The effect of AR matching on difference between predicted and experimental Young's Modulus is relatively small. Difference in yield stress was 33% for the AM representative geometry with AR matching, a slight decrease in comparison to the 41% displayed for the $5 \times 5 \times 5$ (AR of 1:1) lattice. Regardless of FCZ geometry, the numerically derived mechanical properties tend to predict a structure which is stiffer and stronger than those seen experimentally.

4.3.2. FCC lattice FE models

Stress-strain plots for the idealised and AM representative FCC lattices are presented in Fig. 21 and Fig. 22, respectively. Size converging behaviour is displayed similar to the FCZ lattice models with the AM representative FCC lattices again showing a visible more accurate trend toward experiment.

The deformed lattice shape and Von Mises stress distribution for the idealised and AM representative FCC lattice structure (Fig. 23) display regions out-of-plane bending visible in struts located at the structure's edges. This effect is displayed in the upper and lower cell rows with their outer faces bending inwards during compression. The Von-Mises stress distribution indicates more uniform structure-wide plastic deformation especially in nodal regions.

Fig. 24 displays the Young's moduli and yield stress for the simulated idealised and AM representative FCC lattices. Large error with experiment is displayed with the predicted Young's Modulus for both idealised and AM representative cases. As in FCZ models, simulated Young's Modulus is a better predictor of the 2% and 4% unloading moduli.

Matching FCC lattice AR with experiment reduced the difference between the experimentally derived 2% unloading modulus and the simulated Young's Modulus even further. The AR matched AM representative FCC lattice displayed a 23% difference with the experimental 2% unloading modulus. AR matching in FCC lattices improved predicted yield stress with the AM representative geometry exhibiting a 31% difference with experiment.

The simulation of lattice structures without axial struts (those parallel to the loading direction) tend to show a greater agreement with experiment, even without matching the number of unit cell repetitions in experiment. Lattices excluding these struts typically exhibit 'bending-dominated' behaviour, therefore during numerical modelling of AM lattice structures greater attention must be given to lattices in which struts primarily deform due to induced tensile or compressive loads.

FE models of large highly porous lattice structures using a 3D-continuum element approach are extremely computationally expensive,

restricting modelling in this work to a lattice size of $5 \times 5 \times 5$ and $4 \times 4 \times 6$ with associated AR's of 1:1 and 2:3. Therefore, simulations represented a small portion of the experimental parts size used for validation which impacts overall simulation accuracy. AR matching in simulation slightly reduced this impact in both lattice topologies tested, though impacted the 'bending-dominated' FCC lattice more.

The proposed AM representative lattice geometry in this work tended to behave in a more similar manner to experiment when compared with idealised lattices. Comparison of each lattices stress-strain curves (Fig. 17, Fig. 18, Fig. 21 and Fig. 22) shows this effect.

5. Conclusion

Current methods of accommodating AM associated lattice structure defects within CAD and subsequent FE models comprise of: beam element FE models with varying cross-sectional diameters; spheres joined *via* Boolean operations to mimic the strut axis variation; splined-based CAD bodies of revolution to emulate as-manufactured rough strut surfaces; and reduction of strut model diameter to smallest found experimentally. Limitations were observed in the reviewed approaches to defect inclusion, specifically the omission of a varying cross-sectional shape. This work proposes a method of modelling strut defects within lattice structures which overcome the identified deficiencies.

The propose method utilises CAD solid loft techniques governed by a series of elliptical cross-sections to define the AM representative struts geometry. A method of equating an ellipses geometric property with those acquired experimentally was developed. Overall, this research proposes a method for representing typical AM associated geometric defects within CAD models of lattice strut elements more realistically. Also, presenting:

- A method of generating AM representative lattice geometries for FE modelling which is computationally inexpensive in comparison to a CT reconstruction approach.
- Simplistic part design for the analysis of geometric defects in lattice strut elements fabricated *via* additive methods, allowing for examination of different strut diameters and build inclination angles.
- Method which was applied to SLM fabricated Inconel 625, though is applicable for any material or AM specific process.
- Extracted strut geometrical properties can for the basis of a SLM machine-specific database used for numerical modelling of novel lattice topologies, whilst accounting for AM associated defects.
- Geometrical data which can be used to randomly generate AM representative strut CAD models for stochastic finite element modelling (SFEM). This allows insight into various strut fabrication scenarios and therefore critical – or 'worst-case' scenario.

- Allowing insight into the effect of various fabrication scenarios on a struts mechanical properties.
- Phenomenological approach to FE modelling of lattice structures which overcomes deficiencies seen in literature. The model improves accuracy in comparison to idealised lattice structures containing defect free strut geometries.

With the advent of AM, lattice structures have gained increasing interest due to their customisable properties. Though creating accurate AM specific numerical models of these structures remains a challenge due to the scale difference between individual struts which make up the structure and the bulk size of the structure itself. Therefore, future work intends to build upon this AM specific numerical modelling approach to generate fast and accurate beam element models of bulk lattice structures.

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